

Model Architecture Beats Compute: An Energy-Aware Comparison of Four Transformer Encoders for Financial Sentiment

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Abstract—As NLP models grow incredibly large, their increasing energy consumption has come under scrutiny. Many engineers assume that more training energy directly translates into better task performance, but this paper provides a case where this assumption is not necessarily true. Using a modular seven-stage pipeline, the GPU and CPU energy consumption is measured during training and inference runs on transformer-based encoder models (specifically DistilBERT, BERT-base, RoBERTa-base, and FinBERT). The dataset consists of 10,521 Coca-Cola (KO) headlines from the publicly available FNSPID dataset, with datapoints spanning from 2009 to 2023. A one-way analysis of variance (ANOVA) test across the four models shows significant F1 differences ($F = 18.94$, $p = 1.5 \times 10^{-7}$). An analysis of covariance (ANCOVA) test reveals that the underlying model architecture is a leading factor in the weighted F1 result ($p < 10^{-5}$), while training energy provides negligible increases in variance ($p = 0.97$). DistilBERT, the smallest model tested, reaches the highest energy-adjusted F1 value by using roughly half the training energy of larger models. This demonstrates that the choice of model, rather than the compute budget, drives accuracy on this finance-focused task.

I. INTRODUCTION

The standard for AI-usage has shifted towards using the biggest models available for even the smallest of tasks. This has led to push back from experts over concerns of the lack of scalability, sustainability, and cost. Model efficiency is arguably just as important as model accuracy, and the two should be evaluated together when choosing a model [6]. Training a single high-end LLM can emit hundreds of times more CO₂ than five cars over their entire lifetime [7]. Aside from the environmental impact, large models can also take much more time to produce an answer than smaller models. In the context of natural language processing (NLP) on financial data, time is extremely valuable and waiting for an LLM response can cost tangible losses for investors. As happens occasionally, large-pretrained transformers (e.g. BERT-base [2], RoBERTa-base [3], FinBERT [4]) are used for short, trivial classifications. Oftentimes, these model choices are made without quantifiable evidence that the additional compute will lead to improved task performance. These blind decisions can worsen returns due to high latencies and added energy costs. Smaller models like DistilBERT [1], which uses half the layers

of BERT while retaining 97% of its capabilities, are oftentimes dismissed because of a lower observed accuracy, despite the model operating much more efficiently. The assumption that “bigger is better” has propagated into financial NLP research because of the use of general-purpose benchmarks, which do not translate to highly-specified domains.

Existing work on energy-efficient machine learning tend to focus on training-time energy with general benchmarks like GLUE or ImageNet [6], [7], and they tend to neglect the energy cost for inference calls, which albeit can be minuscule compared to training cost, but financial algorithms can execute at extremely large volumes daily and accumulate a large inference energy impact. This paper addresses that gap by treating model architecture as an independent variable,

Existing work on energy-efficient machine learning focuses on training-time energy measured across general-domain benchmarks such as GLUE or ImageNet [6], [7], with relatively small attention on the inference time-energy cost, which can become significant at scale. This project addresses that gap by tracking the time, energy, and accuracy for both inference and training phases across the four NLP models. The resulting analysis will quantify any energy or time differences after partialling out the model architecture.

All in all, this project has three main contributions. First, it presents a seven-stage pipeline that collects and cleans financial data (text and numerical) and can evaluate and compare the energy efficiency any NLP model locally available, where each training and inference run has CPU and GPU energy logging with a 1 Hz resolution. The energy logging is done with Intel RAPL package counters and NVIDIA NVML and integrates the results into a single, time-ordered document. Second, it provides robust analysis on four NLP models (namely DistilBERT [1], BERT-base [2], RoBERTa-base [3], and FinBERT [4]) to examine their energy-accuracy tradeoffs. Each model is fine-tuned on weak-labeled financial headlines for The Coca-Cola Company (KO) over a period from 2009 to 2023 (headlines from the FNSPID corpus [5] and numerical weak-labels from Yahoo Finance/EDGAR). Third, it provides a statistical analysis using analysis of variance (ANOVA) and analysis of covariance (ANCOVA), which show that energy

consumption during the training phase alone explains negligible variance in accuracy when architecture is controlled. The analyses also show that model choice has a significant effect on the task’s F1 result.

II. RELATED WORK

It is important to consider three lines of research for this project: Transformer-based encoder models for sentiment classification, energy-aware machine learning evaluation, and weak supervision for financial sentiment.

For context, all of these models are derivatives of BERT [2], which established the pretrain-then-fine-tune model for transformer-based encoder models on classification tasks. DistilBERT [1] reduces the size of BERT by roughly 40% parameters, allowing it to run 60% faster while maintaining 97% GLUE performance from its predecessor. This makes DistilBERT a representation for general-use lightweight models. FinBERT [4] has the same architecture as BERT, but is pre-trained on billions of financial tokens and is meant to be specialized on finance text. RoBERTa [3] uses bigger batch sizes and more iterations than BERT, resulting in a longer training phase. Prior comparisons of these models do not typically report training and inference energy usage with financial data, and instead focus on general benchmarks like GLUE, SST-2, and Financial Phrase Bank.

Arguments in favor of the model efficiency approach are shaped by works like Schwartz et al. [6], which started the idea of “Green AI” that states energy consumption should be considered alongside accuracy to create a quantifiable “price tag” for models. Strubell et al. [7] is another widely cited paper in the space of energy-accuracy for models, and it provides a robust approach for analyzing NLP energy consumption during the training phase. It’s worth noting that these works do not perform a per-architecture comparative analysis for both the training and inference phases with external CPU and GPU energy measurements. This project intends to fill that gap for the four BERT models.

Snorkel [8] offers an analysis of using weak supervision as a substitute for human annotation. Weak supervision is the use of function-generated labels allowing for the cheap creation of large-scale datasets, and in the context of this project, avoids the bias of human labels on financial sentiment. This project uses weak supervision to generalize the pipeline for any public company, whereas a dataset like the Financial Phrase Bank [9] relies on humans to manually annotate the data, which can be costly and significantly limits the potential size of datasets. The phrase bank serves as a standard benchmark for most financial sentiment studies, including that of FinBERT. Weak labels create a lot more noise than human labels, so this noise decreases the F1 across all models trained on them. However, the same label dataset is used for each model, so the noise does not add bias to the relative model ranking and thus the energy-accuracy comparisons remain valid.

III. METHODOLOGY

A. Pipeline architecture

The pipeline has seven stages, each represented as its own Jupyter Notebook. Each stage saves a local parquet file that the following stage extracts and reads. Stage 1 pulls in ticker-specific financial news headlines from FNSPID as well as daily price data from Yahoo Finance and revenue data from the U.S. Securities and Exchange Commission EDGAR database. Stage 2 aligns and combines the datasets into a single, date-indexed table. Stage 3 fine-tunes and evaluates the four NLP models on the weak-labeled headline dataset as outlined in III-B. Stage 4 is an optional section that adds a linear regression layer. The stage trains and evaluates seven regression models (linear, ridge, lasso, elastic net, XGBoost, LightGBM, CatBoost) on the next-day forward-return prediction, both with and without sentiment features from the Stage 3 models. Stage 5 collects the inference data on a held-out test set of 1,000 headlines. Stages 6 and 7 score the model outputs and create the figures reported here. Since each stage communicates through local parquet files, it’s possible to run any stage independently without needing to re-run upstream stages. Every training and inference call is logged with wall-clock timing and ISO 8601 start/end timestamps, so an externally observed energy log can be integrated post-hoc.

B. Data and labeling

Headlines are pulled from the financial news dataset FNSPID [5], which pairs news articles with stock price data across public U.S. companies. The Coca-Cola Company (ticker: KO) was chosen as the single focus because it has extensive coverage over a long period of time. However, the stock has a very low volatility, making it less prone to fluctuation from news headlines. This can be a subject for future work to evaluate the effect of stock volatility on model performance. The resulting dataset contains 10,521 headlines spanning from 2009-06-22 to 2023-12-16.

Sentiment labels are generated using a weak supervision architecture [8] based on forward price movement. For each headline published on trading day t , the next trading day’s log return r is compared against a volatility-scaled threshold $\tau = 0.50 \times \sigma$, where σ is the 21-day rolling standard deviation of daily log returns. Headlines with $r > \tau$ are labeled positive and $r < -\tau$ are labeled negative with the remainder labeled neutral. By scaling based on volatility, the labeler avoids misreading neutral conditions as positive/negative conditions in high-volatility companies as well as misreading positive/negative as neutral in low-volatility companies. The result distribution on the KO dataset is 28% negative, 44% neutral, and 30% positive.

The dataset is split chronologically with a cutoff at 2021-06-10 in order to make an 80/20 train/test split and skip random shuffling.

C. Energy measurement and statistical analysis

Energy is measured with Intel’s Running Average Power Limit (RAPL) for the CPU and NVIDIA’s built-in Manage-

ment Library for the GPU. Both counters are sampled at 1 Hz and have timestamped records written to a shared CSV file. The logger process runs separately from the notebooks, so the measurements do not affect the GPU usage nor the model throughput.

For each training or inference call, the code logs an ISO 8601 timestamp before/after completion. Then, per-run energy in kWh is calculated by integrating the power samples within the time window. All data was collected on a NVIDIA RTX 4090 running Ubuntu 20.04 and glibc 2.31. The energy and accuracy reporting in this stage follows the algorithms from Schwartz et al. [6] and Strubell et al. [7].

Hyperparameters (maximum sequence length 128, training batch size 16, evaluation batch size 32, epochs 3, learning rate 2×10^{-5} , fp16 enabled) are fixed across all four models, with the only variation in the random seed. Each model is fine-tuned using 10 seeds, so that’s 40 seeds total. Model performance is calculated as the weighted F1 score on the aforementioned held-out test set.

The statistical analysis has four main steps. First, each model’s mean F1 and standard deviation is calculated across the seeds. Second, a one-way ANOVA pass over each model using `scipy.stats.f_oneway` in order to test the global null hypothesis that there’s an equal mean F1 across the models. Third, if the global null hypothesis is reject, then the project applies Tukey’s Honestly Significant Difference (HSD) test (`scipy.stats.tukey_hsd`) to all six pairwise model comparisons in order to identify the specific models that differ. Fourth, ANCOVA using OLS is applied `f1 ~ C(model) + energy_train_kwh` to see if the observed F1 differences are due to changes in training energy rather than the model architecture itself with a Type II sum-of-squares decomposition. The model-effect F-statistic from this ANCOVA is meant to test whether architecture is a significant predictor of F1 after partialling out linear effect of training energy, and the energy-effect F-statistic tests the other way around.

IV. RESULTS

A. Per-model F1 and energy

Table I summarizes the per-model performance across the four transformer-based encoder models, averaged over 10 random seeds and sorted by descending F1. DistilBERT achieves the highest mean weighted F1 (0.335 ± 0.020) with the lowest training time (25.6 s) and energy (0.00224 kWh) than any of the other models. FinBERT and BERT-base have roughly twice the training time and energy costs as well as a lower F1 (0.329 ± 0.018 and 0.327 ± 0.027 respectively). Despite RoBERTa-base consuming the most training time and energy, it resulted in the lowest mean F1 of 0.277 ± 0.009 .

We can observe two patterns. First, the DistilBERT, BERT-base, and FinBERT fall within ± 0.005 F1 of each other. This spread is within the per-seed standard deviation of the models. Second, RoBERTa-base is about 5 points below the rest despite having comparable time and energy to BERT and FinBERT. These results are somewhat unusual, where the cheapest model

TABLE I
MODEL F1 AND TRAINING COST (MEAN OVER 10 SEEDS, $\pm$ STD FOR F1).

Model	F1 (mean $\pm$ std)	Train time (s)	Train energy (kWh)
DistilBERT	0.335 ± 0.020	25.6	0.00224
FinBERT	0.329 ± 0.018	46.7	0.00440
BERT-base	0.327 ± 0.027	46.0	0.00436
RoBERTa-base	0.277 ± 0.009	49.0	0.00459

TABLE II
TUKEY HSD PAIRWISE P-VALUES ON F1 10 SEEDS PER MODEL.

	BERT	DistilBERT	FinBERT	RoBERTa
BERT	—	0.79	1.00	$1.0e-5$
DistilBERT	0.79	—	0.89	$1.0e-6$
FinBERT	1.00	0.89	—	$5.0e-6$
RoBERTa	$1.0e-5$	$1.0e-6$	$5.0e-6$	—

has the best mean F1, and the most expensive model has the worst. This could be because the bigger models are overfitting the noisy train data.

B. Statistical significance

A one-way ANOVA across the four models on the F1 measurements rejects the null hypothesis of equal means by a wide margin ($F(3, 36) = 18.94$, $p = 1.5 \times 10^{-7}$). The models do not share a common mean F1.

To pinpoint the effect’s origin, Tukey’s HSD test was applied, p-value matrix results in Table II. We can see the top three models are statistically insignificant, however RoBERTa-base, differs significantly from each of the other three architectures ($p \leq 1.0 \times 10^{-5}$ in every comparison). Thus the ANOVA is driven by RoBERTa-base’s underperformance, and not by a meaningful spread with the other three models.

To test if the observed variation is explained by training energy rather than model architecture, F1 was modeled jointly on architecture and training energy with ANCOVA using Type II decomposition. We can see the architecture term is highly significant ($F(3, 35) = 13.32$, $p = 5.85 \times 10^{-6}$), whereas the continuous training-energy covariate is statistically insignificant ($F(1, 35) = 0.0019$, $p = 0.97$). This means that architecture, and not training energy, is what drives the F1 differences observed on this task. When architecture is controlled, energy budget alone explains negligible variance.

C. Energy-adjusted F1

Table III reports the ANCOVA-adjusted F1 for each model. In this case, the predicted F1 is evaluated at the dataset’s mean training energy alongside the raw seed mean and delta for adjusting. Training energy entered ANCOVA with a coefficient close to zero, so the magnitude is small, but its direction still gives us insights. DistilBERT’s adjusted F1 increased, showing that its raw mean was achieved at a very small training energy, meanwhile the three expensive models adjusted downwards. (FinBERT from 0.329 to 0.327, BERT-base from 0.327 to 0.325, and RoBERTa-base from 0.277 to 0.275.

TABLE III
ANCOVA-ADJUSTED F1 VS. RAW SEED-MEAN F1 PER MODEL.

Model	Raw F1	Adjusted F1	Δ
DistilBERT	0.335	0.342	+0.007
FinBERT	0.329	0.327	−0.002
BERT-base	0.327	0.325	−0.002
RoBERTa-base	0.277	0.275	−0.003

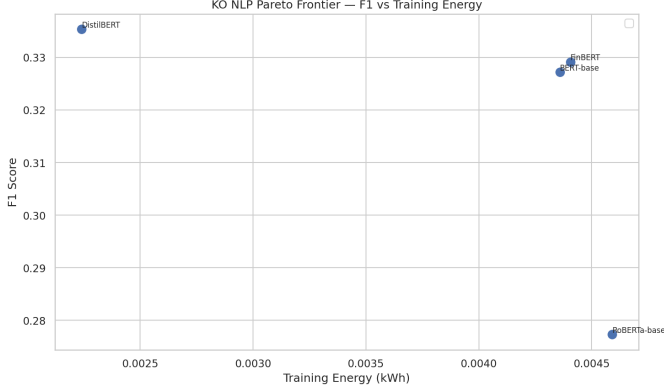


Fig. 1. Pareto frontier of mean F1 vs mean training energy across the four models. Each point is the seed-averaged mean for one model.

After controlling for training energy, DistilBERT’s show much better results compared to BERT and FinBERT. Fig. 1 plots the mean F1 against training energy, and shows that DistilBERT sits much closer to the Pareto frontier, while BERT-base, FinBERT, and RoBERTa-base are dominated by a model that consumes less training energy.

The benchmark in Stage 5 (1,000 headlines, batch size 32) had the same ordering, with DistilBERT at (0.0000256 kWh) inference energy, followed by BERT (0.0000420 kWh), RoBERTa (0.0000528 kWh), and FinBERT (0.0000625 kWh). Thus, the energy ranking was replicated during the inference phase.

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